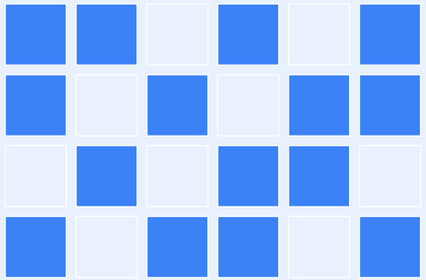
FIGURE 1 | OVERVIEW

Multimodal drafter training

Long visual context, few answer anchors

01 Long context

few answer anchors



C

≫

A

image-conditioned context

question + image tokens

valid anchors

answer positions

padding stays out of the objective

02 Capture once

reuse the image-conditioned record



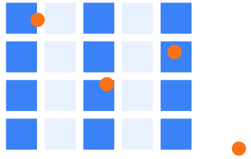
image

question

Frozen target

full image

image-conditioned record



spatial index

C × A

group by context
and anchor count

one target pass

1× target

record reused

many drafter epochs

E1 E2 E3

03 Score by objective

one drafter pass, two selection units



record

Drafter

parallel block

Row-local

score selected rows

only objective-dependent rows

Prefix-coupled

keep each valid block

preserve within-block dependency

Target still reads the full image

optimization lives in record reuse and drafter packing

VelaDraft overview. The full image-conditioned record is captured once; the drafter reuses it and scores only what each objective depends on.

FIGURE 2 | VISUAL EVIDENCE

Visual rows carry proposal signal

Wrong-image replacement hurts more than compacting the visual grid

Visual views

same target trajectory, different drafter rows

Full
reference

Quarter-grid
1 of 4 visual cells

No visual rows
text positions only

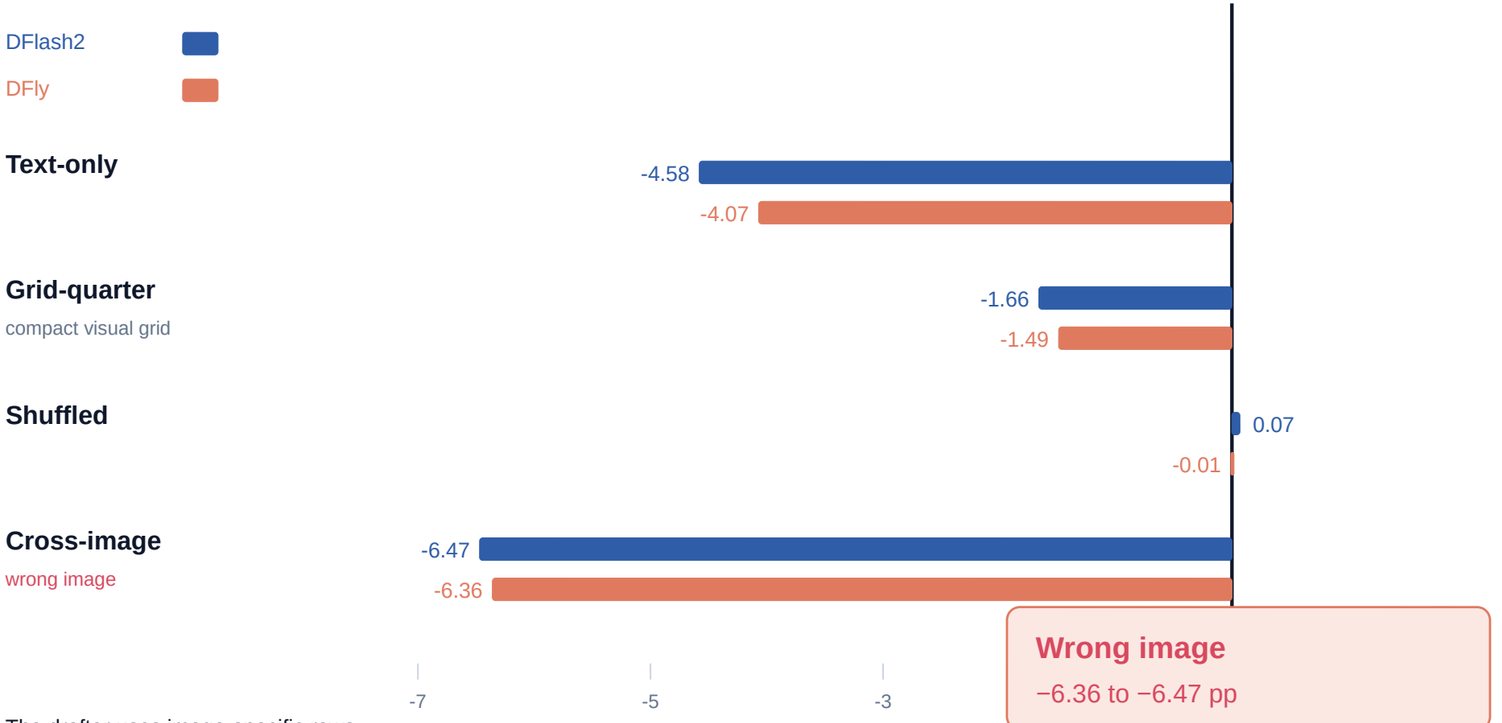
Shuffled
same rows, altered order

Cross-image
wrong image

Full = 0 pp reference

Fixed-trajectory top-1 change vs Full (pp)

negative values indicate fewer matched proposal tokens



The drafter uses image-specific rows

A quarter-grid keeps most of the signal: −1.49 to −1.66 pp

384 image-disjoint requests. Target reads the complete image; only visual rows exposed to the drafter change. Values are fixed-trajectory top-1 changes, not final task accuracy.

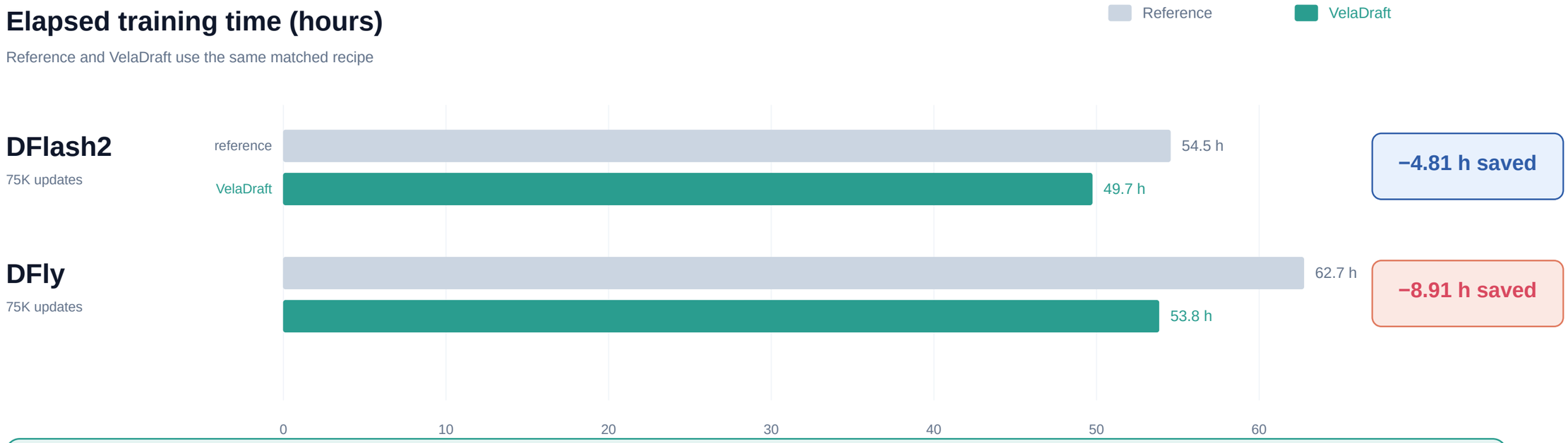
FIGURE 3 | TRAINING COST

Training savings at full scale

Matched 75K-update runs on Qwen3-VL-8B

Elapsed training time (hours)

Reference and VelaDraft use the same matched recipe



Feature reuse

576 → 96

vision-encoder calls across 96 images × 6 passes

480 repeated encodings removed

same image-conditioned records are reused by later epochs

Wall times include training, validation, and checkpoint phases. Reference and optimized runs use matched Qwen3-VL-8B recipes on one A100 80 GB.